Comments for the paper

1. IEEE conference Formatting issues, …… check with the word template (attached)
2. Limit figure captions to 1 line…….. and move the rest of the wording to the description of the figure.
3. Don't your bold or italic font in the paper body text
4. Don't include limitation sub section.
5. Just include a paragraph at the end of that section, just stating the current challenges and future directions.
6. Reduce conclusion to 1 small para
7. Reduce refences to 15
8. Try to Remove all the references before 2021
9. In your latex, keep a separate .bib file, that call at the end of the .tex
10. For tables, use all horizontal and vertical borders , as in the word template.
11. Check 1-to-1 with the formatting of the word template.
12. Index terms, should be keywords
13. There should be some space after the abstract, before starting the keywords.....
14. Figure caption left align
15. Table caption, should follow the same line (should not go to the 2nd line)
16. Use these latest commands for figure/ table/ keywork settings

```
\usepackage{balance}
\usepackage{caption}

\DeclareCaptionFormat{figureformat}{#1#2.  #3}
\captionsetup[figure]{format=figureformat, justification   = raggedright, singlelinecheck = false,font=footnotesize, labelsep=none}

\DeclareCaptionFormat{tableformat}{\textsc{#1}#2.  \textsc{#3}}
\captionsetup[table]{format=tableformat, labelsep=none}


\def\BibTeX{{\rm B\kern-.05em{\sc i\kern-.025em b}\kern-.08em
    T\kern-.1667em\lower.7ex\hbox{E}\kern-.125emX}}

\setlength{\parskip}{6pt}
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to include keywords.....

```
\renewcommand\IEEEkeywordsname{Keywords}

\begin{IEEEkeywords}
write keywords alph order with ,
\end{IEEEkeywords}
```

17. now..... you may see there are several paragraphs, ... where there is only 1 or 2 words, in its last line. In that case, you can edit slightly (eg. use a similar short word to a long word, etc) , and can save 1 line of space (hope you got, what I mean); like wise, you can get more space, if needed

18. Title - use as "Generalized Deep Reinforcement Learning for Traffic Signal Control"

19. Abstract - Don't state the improvement %, … instead state actual performance metric values.
20. Introduction – include terms such as " the contribution of this study are……. where …… is considered as the main novelty"
21. What is the exact novelty to the point? Is that evaluating whether deep reinforcement learning (DRL) traffic signal controllers can generalize to traffic demand levels not encountered during training.
22. Introduction – clearly mention the usefulness of this work.
23. Introduction – clearly mention why you selected these techniques, DQN, Double DQN, PPO and A2C to compare.
24. Introduction – clearly mention, how do you consider Real-City Scenarios, dataset names.
25. Define the reward and performance objectives precisely
26. Are you sure about the correctness of all the equations, and notations?
27. Section IV-B, explicitly distinguish demand-level generalization from route-pattern, temporal, and network-level generalization.
28. In Section IV-D, Justify the shared-policy architecture and its action mapping. Explain how a shared policy handles intersections with different numbers and meanings of green phases. (demand generation and shared-action mapping)
29. How valid phases are represented, whether invalid actions are masked, and how consistent phase semantics are maintained across heterogeneous intersections. An alternative action-mapping strategy can be considered if the current approach introduces action aliasing.
30. State whether invalid actions are masked, whether each intersection has a consistent phase ordering, and how the policy handles phases that are not available at a particular junction.
31. Is it possible to compare the proposed approach with an explicit valid-action mask or another intersection-aware action representation, where feasible.
32. Include a policy-sharing ablation, in Section IV-D. Is it possible to implement the independent-policy baseline for Cologne3 and compare it with the shared-policy configuration under matched training budgets, seeds, hyperparameters, and evaluation conditions. Report the total parameter count, aggregate training cost, and per-intersection performance where appropriate.
33. If the independent-policy experiment cannot be completed, the corresponding contribution should be presented as an unverified hypothesis rather than a demonstrated benefit.
34. In Section III, justify the use of reward-function. Explain why this reward aligns (relationship) with the principal evaluation metric, average total waiting time per vehicle.
35. Specify how accumulated waiting time is computed, whether it includes vehicles waiting to enter the network, and how the reward behaves when vehicles enter, leave, or remain in the network (impact to the reward).
36. Discuss whether the reward difference provides a reliable optimization signal under congestion and whether it can favor short-term reductions in waiting time at the expense of network-wide performance.
37. Report training time, hardware specifications, and computational budgets for every algorithm. Clarify how equal tuning budgets are enforced across value-based and policy-gradient methods, since the same number of episodes does not necessarily imply equivalent optimization effort.
38. Clearly state what you mean by the action timing, minimum-green constraints, yellow transitions, and safety enforcement.
39. Explain how action changes are applied in SUMO, how minimum-green constraints interact with decision timing, and how conflicting movements are prevented from receiving unsafe simultaneous greens.
40. Define the waiting-time metric precisely. The main metric combines time spent halted on the road and time spent waiting to enter the network. Explain how these quantities are obtained from SUMO, whether the same vehicle can contribute to both components, and how vehicles that have not completed their trips by the end of the simulation are handled.
41. Discuss scalability and transferability of the shared policy. Explain whether the intersection identifier is specific to the training network and whether the policy could operate on a new intersection or a new city without retraining.
42. Provide complete hyperparameters
43. Have to include a comparison table with the latest related work. And justify the novelty.
44. Discussion - explain how duplicate trips and shifted departures are handled.

45. Provide a mathematical definition of average total waiting time and explain how incomplete trips are counted.
46. Discuss the limitations of using intersection identifiers and the extent to which the shared policy can transfer to unseen intersections.
47. Is it possible to include complete learning curves with variability across seeds and a fixed-time reference.
48. If you have a github link or any other supportive website for this work you have done (may be a google site like https://sites.google.com/cse.mrt.ac.lk/resqconnect?usp=sharing (if you can prepare) and include the link to support reproducibility.
49. At the end, upload the paper to Stanford Agentic Reviewer https://paperreview.ai/ , and when you get the token, check the review, and refined the paper.